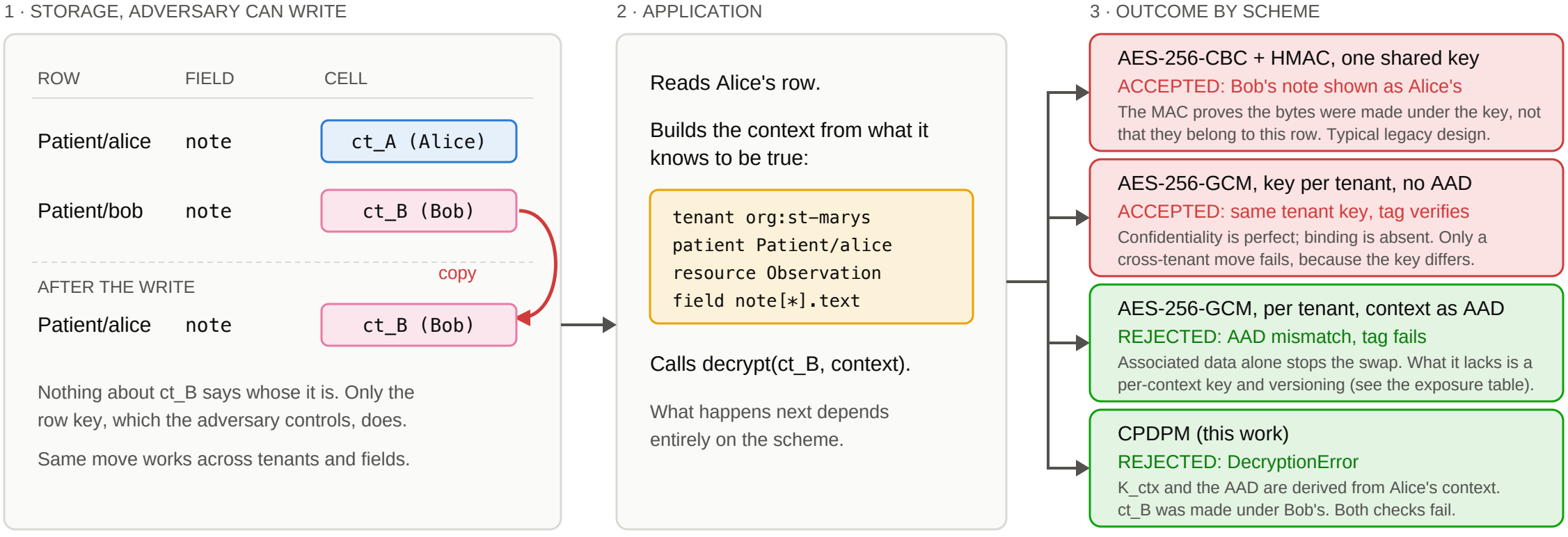
The record-swap attack

An insider with database write access moves Bob's ciphertext into Alice's row. The application, reading Alice's row, decrypts for Alice.



MEASURED, paper/evaluation/record_swap.py

SCHEME	PATIENT SWAP	TENANT MOVE	FIELD MOVE	STALE REPLAY AFTER KEY RETIREMENT
AES-256-CBC + HMAC, one shared key	accepted	accepted	accepted	n/a, no key versioning
AES-256-GCM, key per tenant, no AAD	accepted	rejected	accepted	n/a, no key versioning
AES-256-GCM, key per tenant, context as AAD	rejected	rejected	rejected	n/a, no key versioning
CPDPM (this work)	rejected	rejected	rejected	rejected